Dongchel Shin

PhD Candidate in Mechanical Engineering
Quantum & Precision Measurements Group, Massachusetts Institute of Technology
dongchel@mit.edu · dongchel.github.io · Google Scholar

Research Interests

- Quantum measurement and sensing
- Quantum optomechanics
- Nonlinear optics
- Optical frequency combs
- Precision measurements
- Control and instrumentation

Education

2022 – present **PhD in Mechanical Engineering**, Massachusetts Institute of Technology, USA
Expected completion: May 2027
Thesis: *Quantum Measurement and Control of a Macroscopic Torsional Oscillator*
Advisor: Prof. Vivishek Sudhir

2016 – 2018 **MS in Information Sci. & Tech.**, The University of Tokyo, Japan
Department of Mechano-Informatics
Advisor: Prof. Shoji Takeuchi

2013 – 2016 **BEng**, Tokyo Institute of Technology, Japan
Department of Mechanical and Intelligent Systems Engineering
Summa cum laude; ranked first in the department; completed in 3.5 years

Research Experience

2022 – present **Graduate Research Assistant**, Massachusetts Institute of Technology, USA
Quantum & Precision Measurements Group (Prof. Vivishek Sudhir)

- First PhD student in the group; helped build the laboratory from the ground up
- Quantum-limited measurement and feedback cooling of a centimeter-scale torsional oscillator
- High-Q micro- and nanomechanical resonators
- Cavity-enhanced measurement of the spatial modes of optical fields

2025 **Graduate Research Intern**, NTT Research, USA
PIs: Dr. Myoung-Gyun Suh and Dr. Edwin Ng

- Theory and experiment of quantum-enhanced frequency comb metrology

2018 – 2022 **Research Associate** (alternative military service), KAIST, South Korea
Advisors: Prof. Seung-Woo Kim and Prof. Young-Jin Kim

- Phase-coherent synthesis of ultra-stable terahertz frequencies from optical frequency combs
- Comb-to-comb stabilization over fiber and 1.3-km free-space optical links
- Frequency-comb-referenced plasmonic phase spectroscopy

Publications

Bold: this author; *corresponding author; †equal contribution.

Preprints

[P1] **D.-C. Shin***, E. Ng, M.-G. Suh, V. Sudhir, "Continuum-field quantum optics of frequency comb metrology," arXiv:2605.16702 (2026).

Peer-reviewed journal articles

[J1] M.-G. Suh†,*, D. Dang†, M. Gao†, Y. Jin, **D.-C. Shin**, A. Gupta, B. J. Park, C. Uzundal, B. Hu, W. J. M. Kort-Kamp, H. W. H. Lee*, "Hyperspectral dual-comb compressive ghost imaging with deep learning reconstruction," *Light: Science & Applications* **15**, 380 (2026).

[J2] G. Kang, J. Kim, M.-R. Kim, Y. Lee, **D.-C. Shin**, J. Jeon, H. Kim, D. H. Kim, J. Lee, S. Park, Y.-J. Kim*, "Nanometre-precision terahertz interferometry for battery electrode metrology," *Nature Communications* (2026).

[J3] **D.-C. Shin**†, J. Yang†, D. I. Lee, G. Kang, S.-W. Kim*, Y.-J. Kim*, "Compact, robust all-fiber platform for 1-Hz-linewidth synchronization of optical frequency combs using single comb line extraction," *Optics & Laser Technology* **203**, 115526 (2026).

[J4] **D.-C. Shin**[*], T. M. Hayward, D. Fife, R. Menon, V. Sudhir[*], "Active laser cooling of a centimeter-scale torsional oscillator," *Optica* **12**, 473 (2025).
Featured in *MIT News*: "A cool new way to study gravity."

[J5] D. Fife[*], **D.-C. Shin**, V. Sudhir[*], "Temperature stabilization of a lab space at 10 mK-level over a day," *Review of Scientific Instruments* **95**, 095116 (2024).

[J6] D. A. Nguyen[†], D. H. Kim[†], G. H. Lee, S. Kim, **D.-C. Shin**, J. Park, H.-J. Choi, S.-W. Kim, S. Kim[*], Y.-J. Kim[*], "Real-time monitoring of fast gas dynamics with a single-molecule resolution by frequency-comb-referenced plasmonic phase spectroscopy," *PhotoniX* **5**, 22 (2024).

[J7] **D.-C. Shin**, B. S. Kim, H. Jang, Y.-J. Kim[*], S.-W. Kim[*], "Photonic comb-rooted synthesis of ultra-stable terahertz frequencies," *Nature Communications* **14**, 790 (2023).

[J8] J. Yang, D.-I. Lee, **D.-C. Shin**, J. Lee, B. S. Kim, H. J. Kang, Y.-J. Kim[*], S.-W. Kim[*], "Frequency comb-to-comb stabilization over a 1.3-km free-space atmospheric optical link," *Light: Science & Applications* **11**, 253 (2022).

[J9] **D.-C. Shin**, Y. Morimoto, J. Sawayama, S. Miura, S. Takeuchi[*], "Centrifuge-based step emulsification device for simple and fast generation of monodisperse picoliter droplets," *Sensors and Actuators B: Chemical* **301**, 127164 (2019).

Selected Conference Presentations

[C1] **D.-C. Shin**, T. M. Hayward, D. Fife, R. Menon, V. Sudhir, "Quantum-limited readout and feedback cooling of a centimeter-scale torsional oscillator," *Conference on Lasers and Electro-Optics (CLEO)*, Charlotte, NC, USA (2026).

[C2] **D.-C. Shin**, T. M. Hayward, D. Fife, R. Menon, V. Sudhir, "Active radiation-pressure cooling of a cm-scale torsion pendulum," *APS Global Physics Summit*, Anaheim, CA, USA (2025).

[C3] **D.-C. Shin**, B. S. Kim, H. Jang, Y.-J. Kim, S.-W. Kim, "Terahertz frequency synthesizer referenced to an ultra-stable optical cavity," *Conference on Lasers and Electro-Optics (CLEO)*, San Jose, CA, USA (2022).

Honors and Awards

2024 – 2026	MathWorks Mechanical Engineering Fellowship, MIT (2024–25 and 2025–26)
2024	Ranked first, PhD Qualifying Examination, Department of Mechanical Engineering, MIT
2017	Hatakeyama Award, The Japan Society of Mechanical Engineers (top-ranked graduate of the department)
2016 – 2018	Kwanjeong Educational Foundation Scholarship, South Korea
2013 – 2016	Korea–Japan Joint Government Scholarship

Selected Coursework (MIT)

Optics (2.710), Nonlinear Optics (8.431), Atomic and Optical Physics I (8.421), Quantum Information Science (8.371), Stochastic Systems (2.122), Digital Control Systems (2.171)

Professional Service

Peer review	*Physical Review Letters*, *Physical Review Applied*, *Optics Express*, *Optics Letters*

References

Prof. Vivishek Sudhir, MIT (PhD advisor)
Prof. Nergis Mavalvala, MIT
Prof. George Barbastathis, MIT
Dr. Myoung-Gyun Suh, NTT Research
Prof. Young-Jin Kim, KAIST
Prof. Seung-Woo Kim, KAIST